

D329

Network and Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A software vendor with customers in 40 countries posts a SHA-256 hash next to each installer download so users can check the file. The vendor's website was redesigned last month. Which security goal does the hash support?

- A. Confidentiality, because SHA-256 is encryption
- B. Availability
- C. Integrity
- D. Confidentiality

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An engineer at a defense contractor, who has worked there for 15 years, needs to send a confidential design file to a partner so that only the partner can read it. Both have key pairs issued by the company's certificate authority. Which key encrypts the file?

- A. The partner's public key
- B. The engineer's public key
- C. The partner's private key
- D. The engineer's private key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A manager digitally signs a contract so recipients can confirm it came from her and was not changed. Which key do recipients use to verify the signature?

- A. The manager's public key
- B. The manager's private key
- C. Each recipient's private key
- D. A shared symmetric key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A penetration tester hired by a regional hospital types a string containing OR 1=1 followed by a comment marker into the login form's username field and is logged in without a password. The hospital's site gets 20,000 visits a month. Which attack succeeded?

- A. Cross-site scripting
- B. Brute-force password attack
- C. Denial of service
- D. SQL injection

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A distribution center with 200 workers installs security cameras whose footage the security team reviews after thefts. The cameras cost \$18,000. Which control type are the cameras, based on how they are used?

- A. Corrective
- B. Detective
- C. Preventive, because every visible control stops attacks
- D. Preventive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A SOC analyst at an insurance company confirms that ransomware is encrypting files on one workstation in the claims department. The employee was hired three weeks ago. What should the team do next?

- A. Wipe and reimage the workstation right away.
- B. Write the lessons-learned report.
- C. Contain it by isolating the workstation from the network.
- D. Fire the new employee.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team has removed malware and closed the vulnerability it used. Which phase comes next?

- A.** Containment
- B.** Recovery
- C.** Preparation
- D.** Lessons learned

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A server room at a university, renovated in 2021, holds equipment worth \$300,000. A flood would damage 20% of it, and floods are expected once every 10 years. The room is on the second floor of the library. What is the ALE?

- A.** \$600,000
- B.** \$30,000
- C.** \$60,000
- D.** \$6,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A risk has an ALE of \$24,000 and an SLE of \$8,000. What is the ARO?

- A.** \$16,000
- B.** 3
- C.** \$192,000,000
- D.** 0.33

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A credit union with 11 branches wants true multifactor login for its online banking staff portal. The IT manager, who also coaches youth soccer, is comparing options. Which combination qualifies?

- A.** A password plus a security question
- B.** Two different passwords
- C.** A password plus a code from a hardware token
- D.** A fingerprint plus a face scan

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A system requires a password and then a 6-digit PIN. How many authentication factor categories does it use?

- A.** Six
- B.** Three
- C.** Two
- D.** One

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A forensic analyst arrives at a running, compromised file server at a law firm with 70 attorneys. The server has a 4 TB RAID array and was purchased in 2020. What should the analyst capture first?

- A.** The contents of RAM
- B.** Power off the server, then image the disks
- C.** Archived backup tapes
- D.** A forensic image of the 4 TB array

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A software vendor ~~with customers in 40 countries~~ posts a SHA-256 hash next to each installer download so users can check the file. ~~The vendor's website was redesigned last month.~~ Which security goal does the hash support?

Target word: CIA goal of a hash **Decides:** A hash detects changes to the file.

A	Inverted operation	Hashing is one-way; it is not encryption.
B	Adjacent category	A hash does not keep the download online.
C	Correct	Matching hashes show the file was not altered.
D	Confusion set	A hash does not hide the file's contents.

2. Answer A Concept 2

An engineer at a defense contractor, ~~who has worked there for 15 years,~~ needs to send a confidential design file to a partner so that only the partner can read it. Both have key pairs ~~issued by the company's certificate authority.~~ Which key encrypts the file?

Target word: key for confidentiality **Decides:** Only the partner holds the matching private key.

A	Correct	Only the partner's private key can decrypt it.
B	Adjacent category	Only the engineer could then decrypt it.
C	Inverted operation	The engineer should never have it.
D	Confusion set	That creates a signature anyone with the public key can verify.

3. Answer A Concept 2 · transfer

A manager digitally signs a contract so recipients can confirm it came from her and was not changed. Which key do recipients use to verify the signature?

Target word: key for verifying a signature **Decides:** Signed with the sender's private key, verified with the sender's public key.

A	Correct	Anyone can verify with the signer's public key.
B	Inverted operation	Used to sign; never shared.
C	Confusion set	Used to decrypt messages sent to them, not to verify signatures.
D	Adjacent category	Symmetric keys cannot prove which person signed.

4. Answer D Concept 3

A penetration tester ~~hired by a regional hospital~~ types a string containing OR 1=1 followed by a comment marker into the login form's username field and is logged in without a password. ~~The hospital's site gets 20,000 visits a month.~~ Which attack succeeded?

Target word: name the attack **Decides:** Input changed the database query logic.

A	Confusion set	XSS runs script in a victim's browser, not in the database.
B	Adjacent category	No passwords were guessed.
C	Camouflage	Traffic volume does not describe this attack.
D	Correct	The input altered the server's SQL statement.

5. Answer B Concept 4

A distribution center ~~with 200 workers~~ installs security cameras whose footage the security team reviews after thefts. ~~The cameras cost \$18,000.~~ Which control type are the cameras, based on how they are used?

Target word: control type by function **Decides:** Footage is reviewed after events to identify them.

A	Adjacent category	Corrective controls restore systems after an event.
B	Correct	They identify incidents during or after the fact.
C	Absolute language	Visibility may deter, but it does not prevent.
D	Confusion set	Cameras do not physically stop a theft.

6. Answer C Concept 5

A SOC analyst at an insurance company confirms that ransomware is encrypting files on one workstation in the claims department. ~~The employee was hired three weeks ago.~~ What should the team do next?

Target word: step after analysis **Decides:** Confirmed incident; stop the spread first.

A	Dropped element	Skips containment and may destroy evidence while spread continues.
B	Inverted operation	The final phase, not the next.
C	Correct	Containment limits damage before eradication.
D	Camouflage	Tenure does not decide the next response step.

7. Answer B Concept 5 · transfer

A team has removed malware and closed the vulnerability it used. Which phase comes next?

Target word: phase after eradication **Decides:** Removing the threat is eradication; restoring service follows.

A	Inverted operation	Comes before eradication.
B	Correct	Restore and monitor systems after eradication.
C	Adjacent category	Happens before any incident.
D	Dropped element	Skips recovery.

8. Answer D Concept 6

A server room at a university, ~~renovated in 2021,~~ holds equipment worth \$300,000. A flood would damage 20% of it, and floods are expected once every 10 years. ~~The room is on the second floor of the library.~~ What is the ALE?

Target word: ALE **Decides:** $SLE = \$300,000 \times 0.20 = \$60,000$; $ARO = 0.1$.

A	Inverted operation	Divides by 0.1 instead of multiplying.
B	Wrong stem number	$\$300,000 \times 0.1$ skips the exposure factor.
C	Dropped element	That is the SLE; the ARO is left out.
D	Correct	$\$60,000 \times 0.1 = \$6,000$.

9. Answer B Concept 6 • transfer

A risk has an ALE of \$24,000 and an SLE of \$8,000. What is the ARO?

Target word: solve for ARO **Decides:** $ARO = ALE / SLE$.

A	Adjacent category	Subtracts instead of dividing.
B	Correct	$\$24,000 / \$8,000 = 3$ times a year.
C	Confusion set	Multiplies ALE by SLE.
D	Inverted operation	SLE / ALE flips the ratio.

10. Answer C Concept 7

A credit union ~~with 11 branches~~ wants true multifactor login for its online banking staff portal. The IT manager, ~~who also coaches youth soccer,~~ is comparing options. Which combination qualifies?

Target word: combination that is MFA **Decides:** Factors must come from different categories.

A	Confusion set	Both are something you know.
B	Dropped element	Same category twice; the second factor type is missing.
C	Correct	Something you know plus something you have.
D	Adjacent category	Both are something you are.

11. Answer D Concept 7 • transfer

A system requires a password and then a 6-digit PIN. How many authentication factor categories does it use?

Target word: count factor categories **Decides:** Password and PIN are both something you know.

A	Wrong stem number	The PIN length, not a factor count.
B	Absolute language	Assumes any added step adds a category.
C	Inverted operation	Counts items instead of categories.
D	Correct	Both belong to the knowledge category.

12. Answer A Concept 8

A forensic analyst arrives at a running, compromised file server at a law firm ~~with 70 attorneys.~~ The server has a 4 TB RAID array and ~~was purchased in 2020.~~ What should the analyst capture first?

Target word: first evidence to collect **Decides:** Most volatile first.

A	Correct	Memory is lost when power is removed; it is more volatile than disk.
B	Dropped element	Shutting down destroys the memory evidence.
C	Adjacent category	The least volatile source.
D	Inverted operation	Disk is less volatile and can be captured after memory.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	CIA triad mapping	Encryption protects confidentiality; hashing and digital signatures protect integrity; redundancy, backups, and DDoS protection protect availability.	Saying hashing provides confidentiality.	A control and a question about which security goal it serves.
2	Which key does what	To keep a message confidential, encrypt with the recipient's public key; to sign, the sender uses its own private key and others verify with the sender's public key.	Encrypting with the recipient's private key, or signing with a public key.	Sender and recipient named, and a goal of secrecy or proof of origin.
3	Injection versus cross-site scripting	SQL injection sends database commands through input to the server's database; cross-site scripting injects script that runs in other users' browsers; input validation and parameterized queries or output encoding are the fixes.	Calling any malicious input XSS.	Where the injected code runs: the database or a victim's browser.
4	Control types by function	Preventive stops an event; detective identifies it during or after; corrective restores after; deterrent discourages; compensating substitutes when the primary control is not possible.	Calling a camera preventive.	What the control does at the moment of an incident.
5	Incident response order	Preparation, detection, analysis, containment, eradication, recovery, then lessons learned.	Wiping and rebuilding before containing the spread.	"What should the team do next?" during an incident.
6	ALE calculation	$SLE = \text{asset value} \times \text{exposure factor}$; $ALE = SLE \times ARO$.	Reporting SLE as the annual figure, or dividing by ARO.	Asset value, percentage damaged, and frequency per year.
7	Multifactor authentication	MFA requires factors from different categories: something you know, have, are (and sometimes where you are); two items from the same category are single-factor.	Counting a password plus a security question as two factors.	Two credentials and a question about whether they form MFA.
8	Order of volatility	Collect the most volatile evidence first: CPU registers and cache, then RAM, then swap and temporary files, then disk, then remote logs, then archived backups.	Imaging the hard drive before capturing memory.	A live compromised system and a question about what to collect first.

Course scope (public description): Applications - Network and Security - Applications prepares learners for the CompTIA Security+ certification exam. The course introduces learners to skills in identifying threats, attacks, and

vulnerabilities to organizational security. The learner will also gain skills in designing security solutions for enterprise infrastructures and architectures, as well as in implementing security solutions across hardware, applications, and network services. Learners will be able to execute operations and incident response with tools, policies, forensics, and mitigation techniques, and to analyze information security controls, governance, risk, and compliance.